

Computer vision researcher and CTO/Co-founder of Intellindust, developing reliable and efficient visual intelligence across robust learning, object detection, multimodal generation, and edge AI.

Research impact 30+ top-tier conference and journal publications · 4,800+ Google Scholar citations · h-index 25
Open-source impact 6K+ GitHub stars (updated Aug. 2026)

Experience

- 2023–present **CTO and Co-founder**, *Intellindust*, Shenzhen.
Reliable edge AI for real-time vision applications
- 2021–2023 **Senior Researcher**, *Tencent AI Lab*, Shenzhen.
Multimodal generative intelligence

Education

- 2017–2021 **Ph.D.**, *École des Ponts ParisTech*, Paris.
Thesis: Deep Learning for Near-duplicated Patterns Discovery and Alignment in Artworks
Advisor: Mathieu Aubry
Thesis committee: Patrick Pérez, Alexei A. Efros, Shiry Ginosar, Laurent Heutte, and Yann Gousseau
Impact: *ArtMiner* [5] was featured in *Nature*; the thesis also provided a foundation for the *ERC DISCOVER* project.
- 2013–2017 **Engineering Degree**, *École des Ponts ParisTech*, Paris.
Department of Computer Science and Applied Mathematics
- 2016–2017 **Master’s Degree**, *École Normale Supérieure Paris-Saclay*, Paris.
First-class honors (Mention très bien)
Mathematics, Computer Vision and Machine Learning (Mathématiques, Vision, Apprentissage)
- 2009–2013 **Dual Bachelor’s Degree**, *Université Claude Bernard Lyon & Wuhan University*, Lyon & Wuhan.
First-class honors (Mention très bien)
Mechanics & Physics

Languages

Chinese (native) **French** (fluent; 10 years in France) **English** (fluent)

Skills

- Programming Python (PyTorch), C++, MATLAB, and LaTeX

Academic Service

- Conferences ICML, CVPR, ECCV, NeurIPS, ICCV, ICLR, BMVC, WACV
- Journals IJCV, TPAMI, Pattern Recognition

Awards

- Category C Talent**, Shenzhen Global Talent Recruitment Program for Industry and Information Technology (Artificial Intelligence Track)
- CVPR 2025 2nd Place, Joint Denoising and Demosaicing Challenge NTIRE
- CVPR 2025 2nd Place, Ball Action Spotting Challenge SoccerNet
- CVPR 2025 Honorable Mention, Foundation Few-shot Object Detection Challenge FSOD
- ECCV 2024 1st Place, Open-set Recognition Challenge OOD Generalization in CV Workshop
- CVPR 2024 1st Place, Low-Light Object Detection Challenge Physics-Based Vision Workshop
- CVPR 2024 3rd Place, Low-Light Instance Segmentation Challenge Physics-Based Vision Workshop
- ICML 2022 Outstanding Reviewer ICML Program Committee

Teaching

2021	Digital Humanities Meet Artificial Intelligence	Université Paris Sciences & Lettres
2021	Computer Vision for Mechanics of Materials	École des Ponts ParisTech
2019	Signal Processing and Machine Learning	École des Ponts ParisTech

Student Mentoring

* Denotes a student's first publication at a top-tier conference or journal under my supervision.

2025–2026	Da Li [*] , Shenzhen University (<i>SKEL-CF</i> , ECCV 2026)
2025–2026	Jiping Jin [*] , ShanghaiTech University (<i>SKEL-CF</i> , ECCV 2026)
2025–2026	Chenbin Feng [*] , University of Macau (<i>FSOD-VFM</i> , ICLR 2026)
2024–2025	Haiyang Xie [*] , Wuhan University (<i>SimROD</i> , AAAI 2026)
2023–present	Longfei Liu [*] , Intellindust (<i>EdgeCrafter</i> , TMLR 2026)
2023–2024	Yuting Li [*] , China Three Gorges University (<i>SURE</i> , CVPR 2024)
2022–2023	Jianrong Zhang [*] , Jilin University (<i>T2M-GPT</i> , CVPR 2023)
2022–2023	Yangsong Zhang [*] , Shanghai Jiao Tong University (<i>T2M-GPT</i> , CVPR 2023)
2022	Yangtao Wang [*] , Grenoble Computer Science Laboratory (<i>TokenCut</i> , CVPR 2022)
2022	Wen Guo, Inria RobotLearn
2021–2022	Yingyi Chen [*] , KU Leuven (<i>Sortout</i> , TNNLS 2022)
2020–2021	Oumayma Bounou, École des Ponts ParisTech

Selected Publications

Xi Shen denotes corresponding authorship. Publications are listed in reverse chronological order within each category.

First-author Publications

- [1] **Xi Shen**, Robin Champenois, Shiry Ginosar, Ilaria Pastrolin, Morgane Rousselot, Oumayma Bounou, Tom Monnier, Spyros Gidaris, François Bougard, Pierre-Guillaume Raverdy, Marie-Françoise Limon, Christine Bénévent, Marc Smith, Olivier Poncet, K Bender, Béatrice Joyeux-Prunel, Elizabeth Honig, Alexei A Efros, and Mathieu Aubry. Spatially-consistent feature matching and learning for art collections and watermark recognition. *International Journal of Computer Vision (IJCV)*, 2022. **Project**.
- [2] **Xi Shen**, Alexei A Efros, Armand Joulin, and Mathieu Aubry. Learning co-segmentation by segment swapping for retrieval and discovery. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshop (CVPRW)*, 2022. **Project**.
- [3] **Xi Shen**, Yang Xiao, Shell Xu Hu, Othman Sbair, and Mathieu Aubry. Re-ranking for image retrieval and transductive few-shot classification. In *Neural Information Processing Systems (NeurIPS)*, 2021. **Project**.
- [4] **Xi Shen**, François Darmon, Alexei A Efros, and Mathieu Aubry. RANSAC-Flow: Generic two-stage image alignment. In *European Conference on Computer Vision (ECCV)*, 2020. **Project**.
- [5] **Xi Shen**, Alexei A Efros, and Mathieu Aubry. Discovering visual patterns in art collections with spatially-consistent feature learning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. **Project**.

Corresponding-author Publications

- [6] Longfei Liu, Yongjie Hou, Yang Li, Qirui Wang, Youyang Sha, Yongjun Yu, Yinzhi Wang, Peizhe Ru, Xuanlong Yu, and **Xi Shen**. EdgeCrafter: Compact ViTs for edge dense prediction via task-specialized distillation. *Transactions on Machine Learning Research (TMLR)*, 2026. **Project**.
- [7] Chenbin Feng, Youyang Sha, Longfei Liu, Yongjun Yu, Chiman Vong, Xuanlong Yu, and **Xi Shen**. FSOD-VFM: Few-shot object detection with vision foundation models and graph diffusion. In *International Conference on Learning Representations (ICLR)*, 2026. **Project**.
- [8] Da Li, Jiping Jin, Xuanlong Yu, Wei Liu, Xiaodong Cun, Kai Chen, Rui Fan, Jiangang Kong, and **Xi Shen**. SKEL-CF: Coarse-to-fine biomechanical skeleton and surface mesh recovery. In *European Conference on Computer Vision (ECCV)*, 2026. **Project**.
- [9] Xuanlong Yu, Youyang Sha, Longfei Liu, **Xi Shen**, and Di Yang. A closer look at cross-domain few-shot object detection: Fine-tuning matters and parallel decoder helps. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. **Project**.

- [10] Shihua Huang, Yongjie Hou, Longfei Liu, Xuanlong Yu, and **Xi Shen**. Real-time object detection meets DINOv3. *arXiv*, 2025. [Project](#).
- [11] Shihua Huang, Zhihao Lu, Xiaodong Cun, Yongjun Yu, Xiao Zhou, and **Xi Shen**. DEIM: DETR with improved matching for fast convergence. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. [Project](#).
- [12] Yuting Li, Yingyi Chen, Xuanlong Yu, Dexiong Chen, and **Xi Shen**. SURE: Survey recipes for building reliable and robust deep networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. [Project](#).
- [13] Yuting Li, Dexiong Chen, Tinglong Tang, and **Xi Shen**. HTR-VT: Handwritten text recognition with vision transformer. *Pattern Recognition*, 2024. [Project](#).
- [14] Yangtao Wang, **Xi Shen**, Yuan Yuan, Yuming Du, Maomao Li, Shell Xu Hu, James L Crowley, and Dominique Vaufreydaz. TokenCut: Segmenting objects in images and videos with self-supervised transformer and normalized cut. In *Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2023. [Project](#).
- [15] Jianrong Zhang, Yangsong Zhang, Xiaodong Cun, Shaoli Huang, Yong Zhang, Hongwei Zhao, Hongtao Lu, and **Xi Shen**. T2M-GPT: Generating Human Motion from Textual Descriptions with Discrete Representations. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. [Project](#).
- [16] Yangtao Wang, **Xi Shen**, Shell Xu Hu, Yuan Yuan, James Crowley, and Dominique Vaufreyday. Self-supervised transformers for unsupervised object discovery using normalized cut. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. [Project](#).
- [17] Wen Guo, Yuming Du, **Xi Shen**, Vincent Lepetit, Xavier Alameda-Pineda, and Francesc Moreno-Noguer. Back to mlp: A simple baseline for human motion prediction. In *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2022. [Code](#).
- [18] Yingyi Chen, Shell Xu Hu, **Xi Shen**, Chunrong Ai, and Johan A. K. Suykens. Compressing features for learning with noisy labels. *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2022. [Project](#).
- [19] Yingyi Chen, **Xi Shen**, Yahui Liu, Qinghua Tao, and Johan A. K. Suykens. Jigsaw-vit: Learning jigsaw puzzles in vision transformer. *Pattern Recognition Letters*, 2022. [Project](#).

Collaborative Publications

- [20] Haiyang Xie, **Xi Shen**, Shihua Huang, Qirui Wang, and Zheng Wang. SimROD: A simple baseline for raw object detection with global and local enhancements. In *The Association for the Advancement of Artificial Intelligence (AAAI)*, 2026. [Project](#).
- [21] Yang Li, Youyang Sha, Yinzhi Wang, Timothy Hospedales, **Xi Shen**, Shell Xu Hu, and Xuanlong Yu. From misclassifications to outliers: Joint reliability assessment for classification. *arXiv*, 2026.
- [22] Di Yang, Mahmoud Ali, Xuanlong Yu, **Xi Shen**, Quan Kong, Gianpiero Francesca, and Francois Bremond. MoViE: Broaden your views with human motion for action detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [23] Weihuang Liu, **Xi Shen**, Haolun Li, Xiuli Bi, Bo Liu, Chi-Man Pun, and Xiaodong Cun. Depth-aware test-time training for zero-shot video object segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. [Project](#).
- [24] Lang Rao, Yuan Yuan, **Xi Shen**, Guocan Yu, and Xiaoyuan Chen. Designing nanotheranostics with machine learning. *Nature Nanotechnology*, 2024.
- [25] Yihao Zhi, Xiaodong Cun, Xuelin Chen, **Xi Shen**, Wen Guo, Shaoli Huang, and Shenghua Gao. LivelySpeaker: Towards semantic-aware co-speech gesture generation. In *International Conference on Computer Vision (ICCV)*, 2024. [Project](#).
- [26] Wenxuan Zhang, Xiaodong Cun, Xuan Wang, Yong Zhang, **Xi Shen**, Yu Guo, Ying Shan, and Fei Wang. SadTalker: Learning realistic 3d motion coefficients for stylized audio-driven single image talking face animation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. [Project](#).
- [27] Weihuang Liu, **Xi Shen**, Chi-Man Pun, and Xiaodong Cun. Explicit visual prompting for low-level structure segmentations. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. [Project](#).
- [28] Weihuang Liu, **Xi Shen**, Chi-Man Pun, and Xiaodong Cun. Explicit visual prompting for universal foreground segmentations. In *Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2023. [Project](#).
- [29] Shell Xu Hu, Pablo G Moreno, Yang Xiao, **Xi Shen**, Guillaume Obozinski, Neil D Lawrence, and Andreas Damianou. Empirical bayes transductive meta-learning with synthetic gradients. In *International Conference on Learning Representations (ICLR)*, 2019. [Code](#).
- [30] Yuan Yuan, Yueming Lyu, **Xi Shen**, Ivor W Tsang, and Dit-Yan Yeung. Marginalized average attentional network for weakly-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2019. [Code](#).